**Theorem.** $\mathbb{R}$ *is an ordered field with the Least Upper Bound Property*

*Proof.* **Part 1: $\mathbb{R}$ is an ordered field.**

For $\mathbb{R}$ to be an ordered field, it must be a field, it must be an ordered set, and it must satisfy the following:

(i) For all $a, b$, and $c$ in $\mathbb{R}$, if $a < b$, then $a + c < b + c$.

(ii) If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

These are already a lot, and each of them can be expanded, so for organizational purposes, this will be organized as a nested list.

1. $\mathbb{R}$ is a Field
   (a) Additive
      i. There is an element 0 in $\mathbb{R}$ such that $a + 0 = 0 + a = a$ for all $a$ in $\mathbb{R}$.
      ii. Addition is associative: for any $a, b$, and $c$ in $\mathbb{R}$, $a + (b + c) = (a + b) + c$, so $a + b + c$ is unambiguous.
      iii. Addition is commutative: for all $a, b$ in $\mathbb{R}$, $a + b = b + a$.
      iv. Addition has inverses: for all $a$ in $\mathbb{R}$, there exists some $b$ in $\mathbb{R}$ such that $a + b = 0$. $b$ is denoted as $-a$.
   (b) Multiplicative
      i. There is an element 1 in $\mathbb{R}$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a$ in $\mathbb{R}$.
      ii. Multiplication is associative: for any $a, b$, and $c$ in $\mathbb{R}$, $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, so $abc$ is unambiguous.
      iii. Multiplication is commutative: for all $a, b$ in $\mathbb{R}$, $a \cdot b = b \cdot a$.
      iv. Multiplication has inverses: for all $a$ in $\mathbb{R}$ such that $a \neq 0$, there exists some $b$ in $\mathbb{R}$ such that $a \cdot b = 1$. $b$ is denoted as $a^{-1}$.
   (c) For all $a, b$, and $c$ in $\mathbb{R}$, $a \cdot (b + c) = a \cdot b + a \cdot c$.
2. $\mathbb{R}$ is an ordered set: $\mathbb{R}$ has a relation $<$ such that:
   (a) Exactly one of $a < b$, $a = b$, and $b < a$ is true.
   (b) If $a < b$ and $b < c$, then $a < c$.
3. For all $a, b$, and $c$ in $\mathbb{R}$, if $a < b$, then $a + c < b + c$.

4. If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.

We can go through these one by one. Before we begin, recall the definition of the real numbers, that $\mathbb{R}$ is the set of cuts. A cut is a subset $\alpha$ of $\mathbb{Q}$ satisfying:

1. $\alpha \neq \emptyset$ and $\alpha \neq \mathbb{Q}$.
2. If $p$ is in $\alpha$ and $q < p$, then $q$ is in $\alpha$.
3. If $p$ is in $\alpha$, then there exists some $r$ in $\alpha$ such that $p < r$.

*1.a.i: There is an element* $0$ *in* $\mathbb{R}$ *such that* $a+0=0+a=a$ *for all* $a$ *in* $\mathbb{R}$.

Let's call the additive identity $0^\star$[1]. We can also define the operation $+$ for cuts $\alpha$ and $\beta$.

$$\alpha + \beta = \{r+s : r \in \alpha, s \in \beta\} \tag{1}$$

In this case, we can set $\beta = 0^\star$.

$$\alpha + 0^\star = \{r+s : r \in \alpha, s \in 0^\star\} \tag{2}$$

Apply the associative property to the set definition.

$$\alpha + 0^\star = \{s+r : s \in 0^\star, r \in \alpha\} = 0^\star + \alpha \tag{3}$$

By the definition of the cuts, we can also make some statements about $r$ and $s$.

1. For all $s$ in $0^\star$, $s < 0$.
2. By the reflexive property, for all $r$ in $\alpha$, $r = r$.

Add both of these together.

$$r + s < r + 0 = r \tag{4}$$

Since for all $r$ in $\alpha$ and $s$ in $0^\star$, $r+s<r$. This means every $r+s$ in $\alpha+0^\star$ is in $\alpha$. This means that $\alpha \leq \alpha + 0^\star$.

In order to prove that $\alpha \geq \alpha + 0^\star$, we begin with some term $p$ in $\alpha + 0^\star$, which is equal to the sum $r+s$ of terms $r$ in $\alpha$ and $s$ in $0^\star$. OE$\Delta$

[1] For any $x$, $x^\star = \{p \in \mathbb{Q} : p < x\}$